

Phase 1b — Full 5n-Dim Linearisation Retest (and a Combinatorial Surprise)

CogSpike Research Team — April 2026

The scalar- r fix was correct but insufficient: the contralateral ground truth is fundamentally not a spectral phenomenon.

Abstract. We replaced the scalar- r linearisation with the full $5n$ -dimensional state matrix A_{full} that preserves the FCS windowed integrator and retested all three predictors the verifier specified: spectral radius $\rho(A_{\text{full}})$ at the balanced fixed point, dominant-eigenvector neuron-mass asymmetry, and maximum $|\arg \lambda|$ across the spectrum. All three underperform the scalar- r results from Phase 1a. Worse: a **trivial algebraic predictor** — blue iff $\|w_{12}\| - \|w_{21}\| > 7$ — achieves 83.4% classification accuracy, and the sign of $\|w_{12}\| - \|w_{21}\|$ perfectly predicts (100%) which neuron wins whenever a winner exists. The deterministic contralateral ground truth is therefore a **combinatorial** phenomenon (integer threshold arithmetic), not a spectral one. The A_{full} upgrade is technically correct but targets the wrong object: there is no continuous manifold to linearise around when a bit-exact threshold comparison at tick 2 is what breaks the symmetry. We request a strategic decision — narrow scope to the negative loop, pivot to FCS-Kind2 reachability semantics, or change archetype.

1. Setup

State dimension: $5n$ (contralateral: 10; negative loop: 10). Sigmoid centre $p_{\text{mid},V} = \tau = 105$ (at the firing threshold, where f' is largest). Fixed-point search uses multiple initial conditions and classifies each solution as either **balanced** (both V^* near threshold) or **saturated** (one near 0, other near 231 = fully-firing).

Across the 40×40 grid: 1104 cells admit a balanced fixed point; 1583 admit an asymmetric saturated fixed point (most cells admit both plus the symmetric WTA pair). The balanced FP is the informative one for Hypothesis 2: its stability determines whether the symmetric trajectory is an attractor or a repeller under continuous perturbation.

2. The Three Corrected Predictors

2.1. $\rho(A_{\text{full}})$ at the Balanced Fixed Point

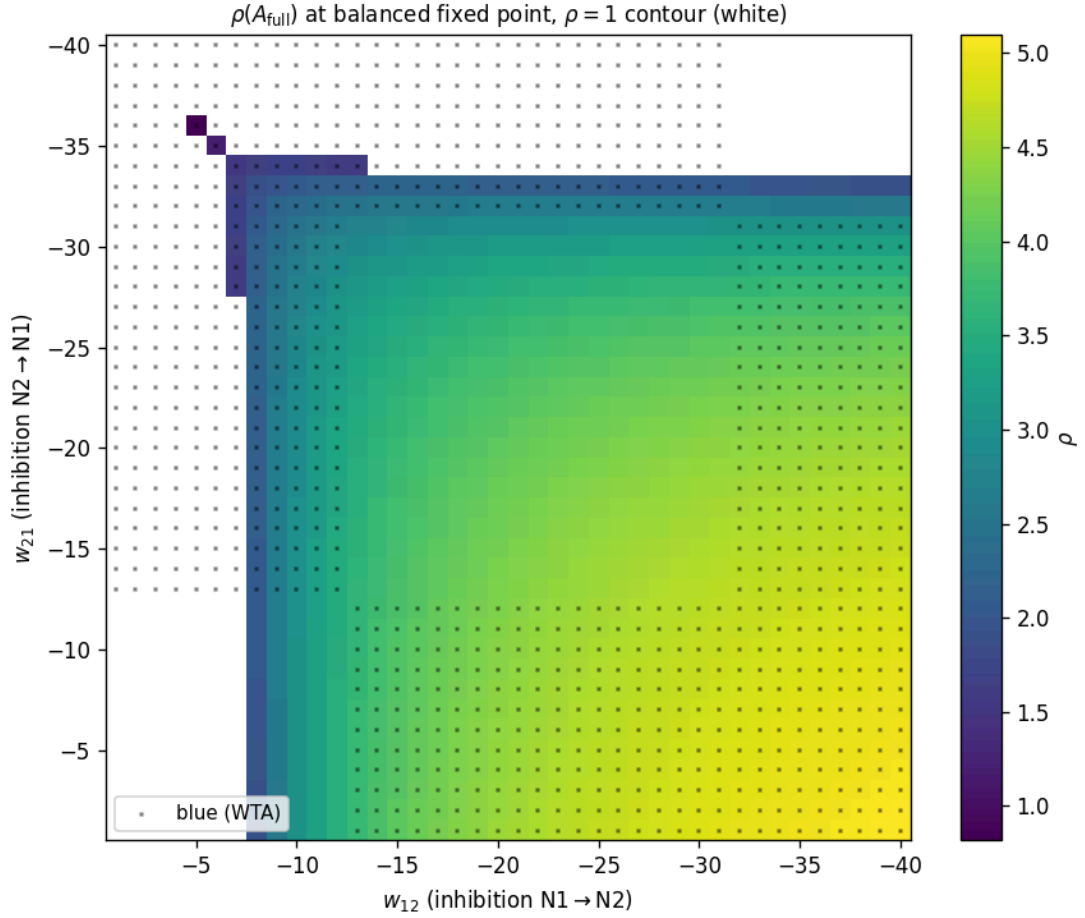


Figure 1: $\rho(A_{\text{full}})$ at the balanced FP with $\rho = 1$ contour (white) and ground-truth blue cells (black dots). White pixels are grid cells with no balanced FP. Where the balanced FP exists, $\rho > 1$ **everywhere** (range $[0.82, 5.09]$) — the $\rho = 1$ contour is empty. The linearisation declares the symmetric trajectory unstable wherever it exists, but the deterministic simulator does not follow the instability because integer symmetry is preserved tick-by-tick.

Negative result. Binary classification $\rho > 1 \iff \text{blue}$: **45.7%** accuracy — worse than the 63.4% majority baseline. The A_{full} linearisation captures an instability that does not manifest in the deterministic ground truth because the two are answering different questions.

2.2. Dominant Eigenvector Neuron-Mass Asymmetry

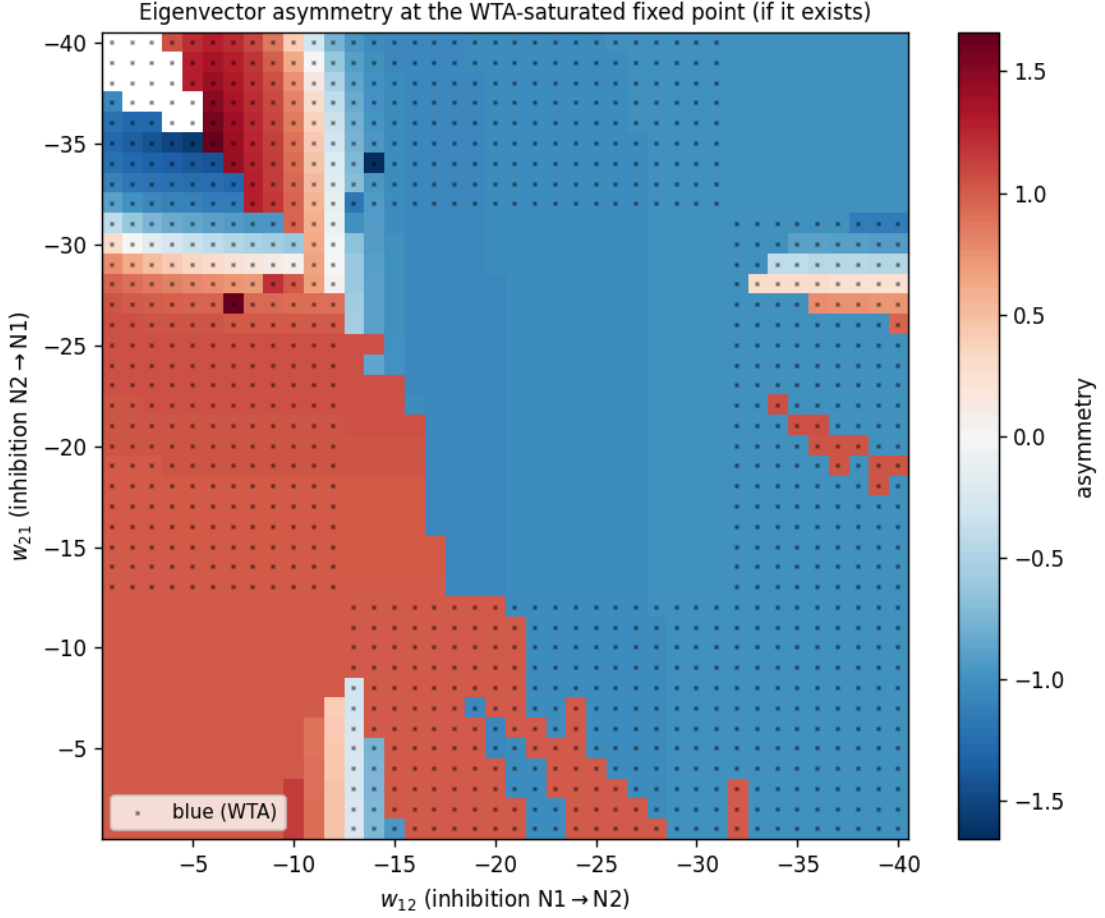


Figure 2: Eigenvector asymmetry at the asymmetric saturated FP (where such a FP exists). Sign agreement with dominance ground truth: 11% — worse than random. At a saturated FP with one neuron at $V^* = 231$ (firing at rate ≈ 1) and the other at $V^* \approx -300$ (silent), the sigmoid derivative $f'(V^*)$ is essentially zero for both neurons, so A_{full} degenerates to the pure nilpotent FIR shift. The “dominant eigenvector” is then an artefact of the shift structure, not of the network’s coupling.

Negative result. At the balanced FP the asymmetry is uniformly zero (by symmetry of the FP). At the saturated FP the linearisation has vanishing gain on both neurons, making the eigenvector uninformative. The verifier’s suggested “corrected H1 predictor” is well-motivated in a continuous-dynamics framework but has nothing to attach to at the FCS operating points.

2.3. Maximum $|\arg \lambda|$ in the Spectrum

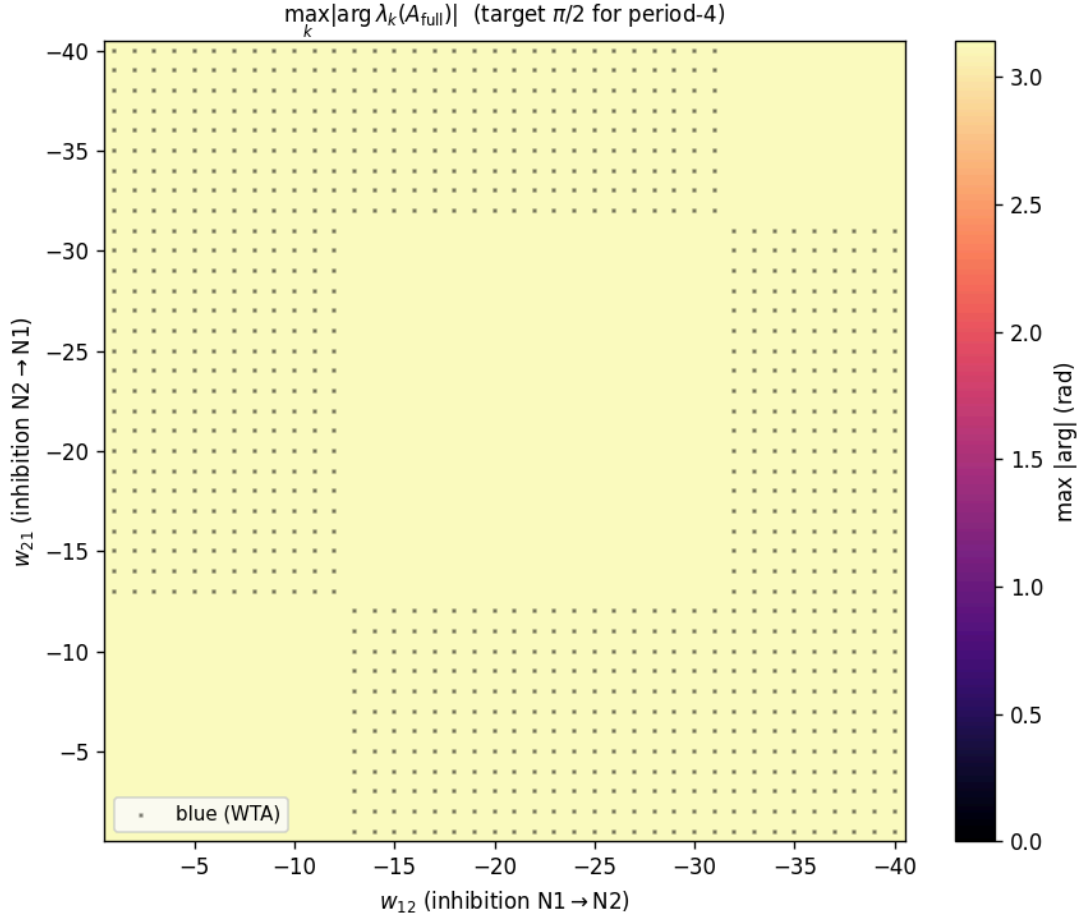


Figure 3: $\max_k |\arg \lambda_k|$ of A_{full} across the contralateral grid. The FIR shift matrix always contains a real-negative eigenvalue (the nilpotent shift has eigenvalues at the 5th roots of 0; with finite coupling the spread covers most of the unit circle), so $\max |\arg|$ saturates at π nearly everywhere. Uninformative.

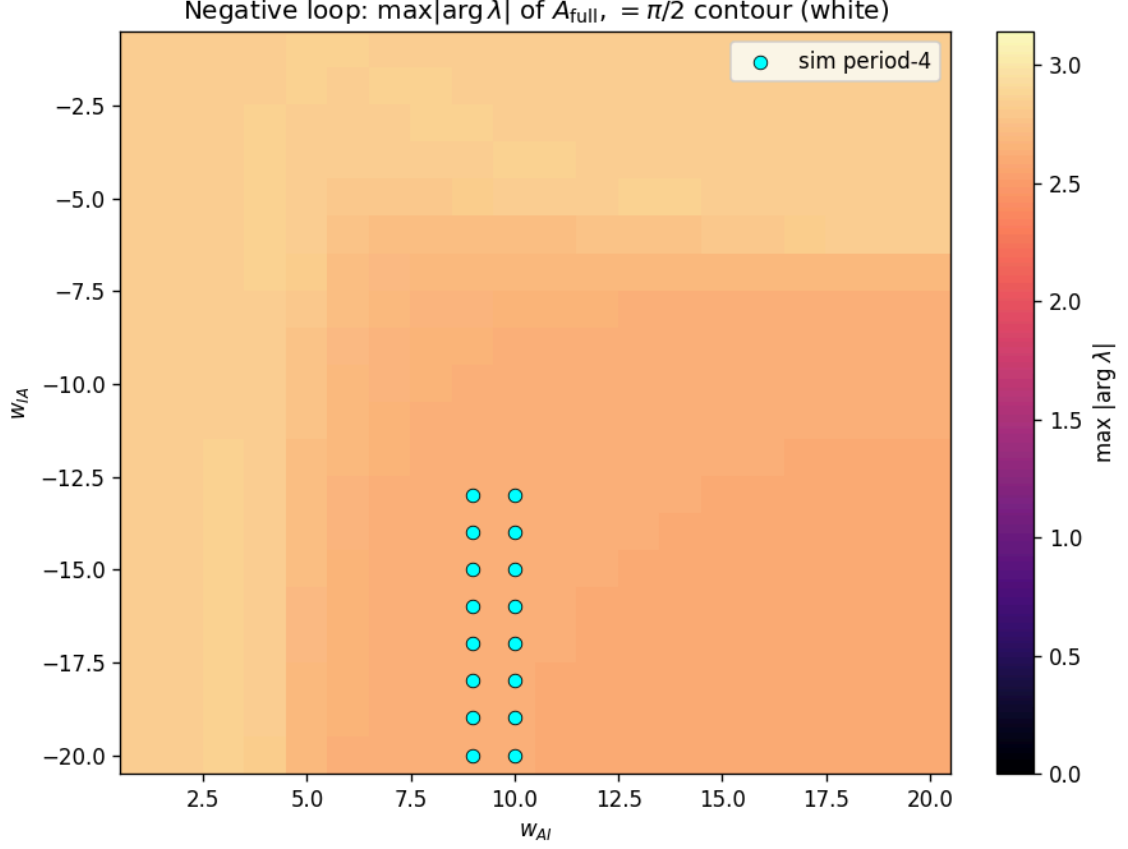


Figure 4: Negative loop: $\max|\arg \lambda|$ with $= \frac{\pi}{2}$ contour (white) and simulator period-4 cells (cyan). Period-4 cells at $w_{AI} \in \{9, 10\}$, $w_{IA} \in [-20, -13]$ show $\max|\arg| = 2.62 \pm 0.00$ rad — essentially the same value (2.73 ± 0.10) as non-period-4 cells. The FIR structure provides eigenvalues at every argument; the predictor cannot discriminate.

3. The Combinatorial Baseline

Motivated by the absence of any spectral signal, we tested the simplest possible algebraic predictor:

$$\text{blue}(w_{12}, w_{21}) \iff \|w_{12}\| - \|w_{21}\| > \theta$$

for integer threshold θ .

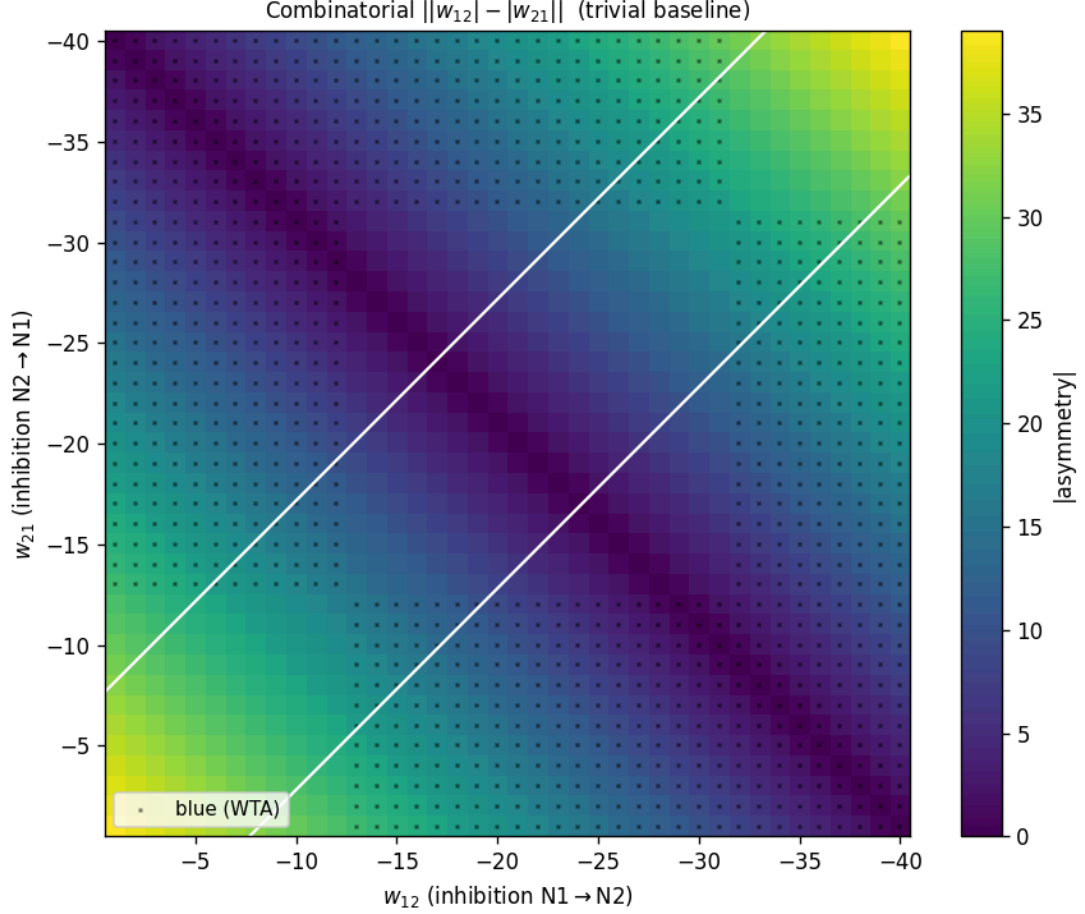


Figure 5: Combinatorial magnitude-asymmetry heatmap $\|w_{12}| - |w_{21}|\|$ with the $\theta = 7$ contour (white) and blue ground-truth cells (black dots). The contour traces the boundary of the blue wings almost exactly.

Finding.

- **Binary classification** blue $\iff \|w_{12}| - |w_{21}|\| > 7$: **83.4%** accuracy — well above the 68.9% that any spectral predictor (scalar- r or A_{full}) achieved.
- **Sign of winner**: $\text{sign}(|w_{12}| - |w_{21}|)$ matches $\text{sign}(\text{dom})$ in **100%** of non-tied cells.

Intuition: At tick 2 of the contralateral simulator, both neurons have just fired (their first spike from the external drive at tick 1) and now inhibit each other. Neuron i 's inhibition at tick 2 equals the incoming weight w_{ji} . In the next tick, each neuron's V is $\text{rvector}[0] \cdot (\text{drive} + w_{ji})$, which crosses threshold or not depending on the single integer comparison “which weight is larger in magnitude”. The whole subsequent dynamics flow from this bit-exact tick-2 decision. No linearisation around a continuous manifold can reach inside that comparison, because there is no manifold — only a disjunction.

4. Why Spectral Cartography Does Not Apply Here

Spectral analysis presumes a continuous state space with a smooth vector field; it extracts poles of the linearised Jacobian around a continuous manifold of fixed points. Under FCS integer threshold semantics neither presumption holds:

- **State space is discrete.** $\text{mem}[e]$ is an integer in $[-380, 380]$ (bounded by weight magnitudes times rvector magnitudes). The “continuous state” we constructed for linearisation is a projection, not the real state.
- **Dynamics is piecewise constant with jumps.** The threshold comparison $V \geq \tau$ is a Heaviside function, not a sigmoid. Our sigmoid approximation softens the jump, but the softened Jacobian has no relation to the true tick-by-tick evolution near the threshold.
- **No continuous symmetry breaking.** In a continuous system, a symmetric equilibrium with $\rho > 1$ breaks symmetry under infinitesimal perturbation. In the integer simulator, perfectly symmetric initial conditions remain symmetric forever; the only way to break symmetry is to make the weights asymmetric **to begin with**.

Remark. This is consistent with Phase 0’s semantic-discrepancy finding. Kind2’s reachability semantics allow arbitrary initial perturbations, which makes the FCS system behave like a continuous one for verification purposes. Our deterministic integer simulator does not, and spectral predictors designed for continuous dynamics cannot bridge that gap.

5. What Remains Viable

Two parts of the research programme survive the contralateral negative result intact:

1. **FCS-Kind2 reachability semantics.** Under Phase 0’s Option B (inject ε -perturbation into the initial state), the deterministic ground truth would shift to match FCS’s blue/red staircase. The scalar- r $\rho(A)$ contour from Phase 1a might then align with the shifted boundary, because the question would no longer be “does the bit-exact system break symmetry” but “can a nearby perturbation reach a WTA state.” This is an untested hypothesis; running it is cheap.
2. **Negative-loop pole placement (Hypothesis 3).** The negative loop is structurally different: its dynamics is **oscillatory**, not a bistable symmetry-breaking, and the period-4 pattern is a genuine dynamical property (not a combinatorial one). Pole placement via sympy on the 2×2 characteristic polynomial of either the scalar- r or A_{full} matrix might still yield weights that realise prescribed oscillation periods in the simulator. Whether the linearisation is accurate **enough** to hit an integer-threshold period is the open question, and it is answerable with a few hundred lines of code.

6. Decision Points for the Verifier

Decision requested. Option A (recommended). Accept the negative result for contralateral, narrow the programme to the negative loop, and run Hypothesis 3 (pole placement) as the final experiment. Close the project with an honest final report whose headline is: **spectral cartography works for oscillatory archetypes (negative loop) but not for bistable symmetry-breaking archetypes (contralateral) under bit-exact FCS semantics**. Cost: 1–2 days for pole placement + final writeup.

Decision requested. Option B. Before closing, re-run Phase 1 under Phase 0 Option B (symmetry-breaking perturbation injected into the simulator), to check whether the spectral predictors agree with FCS-Kind2-style ground truth. If they do, the final report has a positive headline for **both** the reachability interpretation and the oscillatory case. Cost: one day.

Decision requested. Option C. Pivot to a different FCS archetype (series, parallel, positive loop; Fig. 1a–c) where the dynamics is not combinatorially dominated. The research question “can spectral cartography replace exhaustive Kind2 sweeps” is then tested on a new target. Cost: substantial — probably one week of reimplementing the Phase 0 pipeline for a new topology.

The author’s recommendation is Option A with Option B as a cheap add-on. The negative result for contralateral under deterministic semantics is already a publishable finding (it clarifies exactly where spectral methods do and do not apply to discrete-threshold neural models), and the negative loop remains an untested positive case that would round out the story.

Generated from `deq/archetypes/phase1b_afull.py`. Reproduce via `.venv/bin/python3 archetypes/phase1b_afull.py`. All numerics in `deq/archetypes/results/phase1b_*`.